

PDAC Data Audit

Dataset Inventory and Integration Assessment for Supervised Bayesian Matrix Factorization

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1 Executive Summary

This document provides a structured inventory of the PDAC (Pancreatic Ductal Adenocarcinoma) datasets available for use with the supervised Bayesian matrix factorization model implemented in `multiomicsGEP`. Seven datasets are available, spanning RNA-seq, microarray, and proteomics platforms. Each dataset includes a gene expression matrix, survival data (time-to-event with censoring), and CAF (cancer-associated fibroblast) subtype annotations.

All seven datasets are compatible with `fit_supervised_mf_modular()` after standardised pre-processing via the `load_data_internal()` function. With top-5000 variance-filtered genes, the estimated total compute time for all datasets is approximately one minute.

2 Data Location and File Structure

2.1 Source Directory

```
~/OneDrive/UNC Dissertation (Liu)/PDAC_data/  
├── load_data_internal.R      ← Standardised loader function  
└── original/                ← Raw data files (7 datasets)
```

2.2 Per-Dataset File Convention

Each dataset {name} has four associated files:

File	Contents	Format
{name}.rds	Expression matrix + sample metadata + feature annotations	R list
{name}.survival_data.rds	Survival times, event indicators, whitelist flags	R data.frame
{name}_subtype.csv	CAF subtype classifications (SCISSORS, PurlST, etc.)	CSV
{name}.caf_subtype.rds	Same subtype data in RDS format	R object

Additionally, `cmbSubtypes.RData` is present as an ancillary reference file for combined subtype analysis. It is not required for the matrix factorization pipeline.

3 Dataset Inventory

Table 2: Summary of available PDAC datasets

Dataset	Platform	Genes	Samples_Total	Samples_Valid	Events	Censored	Censoring_Rate	Time_Range	Log2_Applied
TCGA_PAAD	RNA-seq	20531	181	144	75	69	47.9	0.1 – 76.2	Yes
CPTAC	Proteomics	28057	140	134	68	66	49.3	0.0 – 45.5	No
Dijk	RNA-seq	49677	90	90	81	9	10.0	2.0 – 119.8	Yes
Moffitt_GEO_array	Microarray	19749	357	123	83	40	32.5	1.0 – 59.0	No
PACA_AU_array	Microarray	17265	131	88	54	34	38.6	0.3 – 47.0	No
PACA_AU_seq	RNA-seq	56206	92	69	41	28	40.6	0.3 – 50.4	Yes
Puleo_array	Microarray	20087	309	288	181	107	37.2	1.1 – 146.4	No

Notes:

- *Samples_Valid* reflects samples retained after survival validation (time > 0, non-missing time and event) and dataset-specific filtering applied by `load_data_internal()`.
- *Log2_Applied* indicates whether `load_data_internal()` applies a $\log_2(x + 1)$ transformation to the expression matrix. Datasets already on a log or normalised scale (microarray, proteomics) are not re-transformed.
- *Time_Range* is in months for all datasets.

4 Per-Dataset Profiles

4.1 TCGA_PAAD

Source: The Cancer Genome Atlas — Pancreatic Adenocarcinoma project.

Platform: RNA-seq (raw counts → log2-transformed by loader).

Dimensions: 20,531 genes × 181 samples (144 after filtering).

Expression range (raw): 0 – 6,699,497 (counts); after $\log_2(x + 1)$: 0 – 22.7.

Feature annotation: `featInfo` is a `data.frame` with columns `SYMBOL` and `ENTREZID`. Gene symbols are set as row names by the loader.

Sample filtering: Retains samples with non-missing tumour grade (`!is.na(Grade)`).

Survival: 75 events / 69 censored (47.9% censoring). Follow-up range: 0.1 – 76.2 months.

Sample metadata: Rich clinical annotation (75 columns) including tumour grade, purity estimates, KRAS mutation status, molecular subtype classifications (Moffitt, Collisson, Bailey), staging, and treatment history.

Additional objects in .rds: `decoderWeights`, `subTME` (tumour microenvironment scores).

Suitability: Ideal starting dataset — widely studied, moderate size, comprehensive annotation. Recommended as the smoke-test target.

4.2 CPTAC

Source: Clinical Proteomic Tumor Analysis Consortium — PDAC proteomics.

Platform: Mass spectrometry proteomics (already normalised; not log-transformed).

Dimensions: 28,057 proteins × 140 samples (134 after filtering).

Expression range: 0 – 24.8 (normalised abundance, likely log2-transformed upstream).

Feature annotation: `featInfo` is a list of length 1 (minimal annotation).

Sample filtering: Retains samples with `histology_diagnosis == "PDAC"`.

Survival: 68 events / 66 censored (49.3%). Follow-up: 0 – 45.5 months.

Note: This is a *proteomics* dataset, not transcriptomics. The supervised MF model was derived for gene expression, but the mathematical framework ($Y = LF' + E$ with Cox PH) applies to any continuous omics matrix. Results should be interpreted in the proteomics context (protein abundance programs rather than gene expression programs). Features are proteins, not genes — top-feature tables will report protein identifiers rather than gene symbols.

4.3 Dijk

Source: Dijk et al. — PDAC RNA-seq cohort.

Platform: RNA-seq (raw counts □ log2-transformed by loader).

Dimensions: 49,677 genes × 90 samples (90 after filtering — all retained).

Expression range (raw): 0 – 89,627 (counts); after $\log_2(x + 1)$: 0 – 16.5.

Feature annotation: `featInfo` is a `data.frame` with `SYMBOL`, `GeneID`, `Biotype` — the richest feature annotation among the datasets.

Sample filtering: No filtering applied (all 90 samples have valid survival).

Survival: 81 events / 9 censored (10.0% censoring — very low). Follow-up: 2.0 – 119.8 months.

Note: The very low censoring rate (10%) means nearly all patients experienced the event. This is unusual and may reflect the study design or patient population. High event rates provide strong signal for Cox model estimation but reduce generalisability.

Sample metadata: Includes age, sex, lymph node metastasis ratio, organism part. The .rds also contains a `cnt` object (raw count matrix) in addition to `ex`.

4.4 Moffitt_GEO_array

Source: Moffitt et al. — GEO microarray cohort.

Platform: Microarray (already normalised; not log-transformed by loader).

Dimensions: 19,749 genes × 357 samples (123 after filtering).

Expression range: 0 – 13.1 (log2-scale microarray intensities).

Feature annotation: featInfo is a data.frame with column SYMBOL.

Sample filtering: Retains primary tumour specimens only (specimen_type == "Primary"). The large reduction from 357 to 123 samples reflects exclusion of non-primary specimens (metastases, adjacent normal, stroma).

Survival: 83 events / 40 censored (32.5%). Follow-up: 1.0 – 59.0 months.

Sample metadata: Includes Moffitt tumour/stroma classifications, GEO accession IDs. The .rds also contains decoderWeights.

4.5 PACA_AU_array

Source: Pancreatic Cancer — Australia (ICGC) — microarray platform.

Platform: Microarray (already normalised; not log-transformed by loader).

Dimensions: 47,265 probes × 131 samples (88 after filtering).

Expression range: 1.0 – 14.1 (log2-scale microarray intensities).

Feature annotation: featInfo is a data.frame with PROBEID, ENTREZID, ENSEMBL, SYMBOL. Multiple probes may map to the same gene — the loader resolves this by taking the median across duplicated gene symbols.

Sample filtering: Retains primary PDAC tumours only (Sample.type == "Primary tumour" AND HistoSubtype == "Pancreatic Ductal Adenocarcinoma"). Also aggregates duplicate genes.

Survival: 54 events / 34 censored (38.6%). Follow-up: 0.3 – 47.0 months.

Sample metadata: ICGC identifiers, qpure tumour purity scores, histological subtypes.

4.6 PACA_AU_seq

Source: Pancreatic Cancer — Australia (ICGC) — RNA-seq platform.

Platform: RNA-seq (raw counts → log2-transformed by loader).

Dimensions: 56,206 genes × 92 samples (69 after filtering).

Expression range (raw): 0 – 1,224,510 (counts); after $\log_2(x + 1)$: 0 – 20.2.

Feature annotation: `featInfo` is a `data.frame` with ENSEMBL, ENTREZID, SYMBOL. Duplicate genes aggregated by median.

Sample filtering: Same criteria as PACA_AU_array (primary PDAC tumours). Column names are suffixed with `_seq` to distinguish from the array cohort.

Survival: 41 events / 28 censored (40.6%). Follow-up: 0.3 – 50.4 months.

Note: This is the same patient cohort as PACA_AU_array but profiled on a different platform. Paired analysis could compare factor recovery across platforms.

4.7 Puleo_array

Source: Puleo et al. — large European PDAC microarray cohort.

Platform: Microarray (already normalised; not log-transformed by loader).

Dimensions: 20,087 genes × 309 samples (288 after filtering).

Expression range: 0.3 – 14.3 (log2-scale microarray intensities).

Feature annotation: `featInfo` is a `data.frame` with column SYMBOL.

Sample filtering: All samples retained (no additional filtering beyond survival validation).

Survival: 181 events / 107 censored (37.2%). Follow-up: 1.1 – 146.4 months.

Note: Largest dataset by valid sample count (n=288). Longest follow-up (>12 years). Provides the most statistical power for survival modelling. The rich sample metadata includes TNM grading, resection margin status, sex, and sampling site.

5 The `load_data_internal()` Function

5.1 Location

~/OneDrive/UNC Dissertation (Liu)/PDAC_data/load_data_internal.R

5.2 Interface

```
load_data_internal(dataname)
# Returns: list(ex, sampInfo, featInfo, dataname)
```

Input: A dataset name string matching one of the supported datasets.

Output:

Field	Type	Description
ex	numeric matrix	Gene expression (genes $\times$ samples)
sampInfo	data.frame	Sample metadata with ID, time, event, keep columns
featInfo	character vector	Gene/feature symbols (row names of ex)
dataname	character	Dataset name tag

5.3 Processing Pipeline

The function executes the following steps for each dataset:

1. **Load raw data:** Read `.rds` (expression + metadata), `.survival_data.rds` (survival), and `_subtype.csv` (CAF annotations) from `data/original/`.
2. **Dataset-specific preprocessing:**

Table 4: Dataset-specific preprocessing applied by `load_data_internal()`

Dataset	Sample_Filter	Expression_Transform
Dijk	None (all retained)	$\log_2(x + 1)$
CPTAC	<code>histology == PDAC</code>	None
TCGA_PAAD	<code>!is.na(Grade)</code>	$\log_2(x + 1)$; rownames set to SYMBOL
Moffitt_GEO_array	<code>specimen_type == Primary</code>	None
PACA_AU_array	<code>Primary tumour + PDAC histology</code>	Median-aggregate duplicate genes; remove NA symbols
PACA_AU_seq	<code>Primary tumour + PDAC histology</code>	Median-aggregate duplicates; $\log_2(x + 1)$; suffix <code>_seq</code> to IDs
Puleo_array	None (all retained)	None

3. **Survival validation:** Retain samples where `time > 0`, `!is.na(time)`, and `!is.na(event)`. The intersection of dataset-specific and survival-valid sample indices defines the final keep flag.
4. **Sample metadata assembly:** Left-join survival and CAF subtype data. Add keep indicator column. Validate event indicators are 0/1, times are positive.

5. **Data integrity checks:** Verify expression matrix has column names, sampInfo has non-missing ID, no duplicate sample IDs, expression columns exactly match sampInfo\$ID (re-order if needed).

5.4 Datasets Referenced But Not Present

The loader supports four additional datasets — Grunwald, Hayashi, Linehan, and Olive — but their .rds files are **not present** in the original/ directory. These are skipped in the current analysis.

6 Integration with Supervised Bayesian MF

6.1 Input Contract

fit_supervised_mf_modular(Y, time, status, K, ...) expects:

Parameter	Type	Shape	Description
Y	numeric matrix	n × p	Patients (rows) × genes (columns)
time	numeric vector	n	Survival/censoring time
status	integer vector	n	Event indicator (1 = event, 0 = censored)

6.2 Required Transformation

load_data_internal() returns ex as **genes × samples** (rows = genes, columns = samples). The model expects **patients × genes**. Therefore:

```
data    <- load_data_internal("TCGA_PAAD")
keeps   <- which(data$sampInfo$keep == 1)
Y       <- t(data$ex[, keeps])      # transpose: genes×samples → patients×genes
time    <- data$sampInfo$time[keeps]
status  <- as.integer(data$sampInfo$event[keeps])
```

6.3 Gene Filtering Recommendation

With 20K–56K genes per dataset, variance-based filtering to the top 5,000 most variable genes is recommended as the default. This is standard practice in genomics dimensionality reduction: it removes low-variance housekeeping genes that contribute noise without informative signal, and substantially reduces runtime. The filter should be configurable (including the option to disable it) to allow sensitivity analysis.

6.4 Estimated Compute Time

Table 6: Estimated runtime per dataset (K=5, max_iter=100, top 5,000 genes)

Dataset	n	p_filtered	Est_Seconds
Dijk	90	5000	4
CPTAC	134	5000	6
TCGA_PAAD	144	5000	6
Moffitt_GEO_array	123	5000	5
PACA_AU_array	88	5000	4
PACA_AU_seq	69	5000	3
Puleo_array	288	5000	12
Total	NA	NA	40

Benchmarked against the synthetic simulation (n=250, p=1000, K=5) which completes in ~2.6 seconds on a 2024 MacBook Pro. Runtime scales linearly with $n \times p \times K$.

7 Recommendations

1. **Primary smoke-test dataset:** TCGA_PAAD — well-studied, moderate dimensions, rich clinical annotation, balanced censoring.
2. **Run all seven datasets** sequentially in a single script invocation. Total compute is approximately one minute with top-5000 gene filtering.
3. **Cross-platform comparisons:** PACA_AU_array and PACA_AU_seq share the same patient cohort on different platforms — factor recovery comparison is a natural future analysis.
4. **CPTAC caveat:** This is a proteomics dataset. Results should be interpreted as protein abundance programs rather than gene expression programs.
5. **Dijk caveat:** Very low censoring (10%) — nearly all patients experienced the event. Cox model estimation will be strong but generalisability may be limited.
6. **Gene filtering:** Default to top 5,000 by variance, with a configurable top_n_genes parameter that accepts NULL to disable filtering.
7. **Longleaf future flag:** Data may also be accessible at /proj/rashidlab/data/PDAC on the UNC Longleaf HPC cluster. A PDAC_DATA_R00T environment variable should be added in a future update to support both local and cluster paths.